ECE 310 Fall 2026
Lecture 20: Discrete Fourier Transform

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Learning Objectives

After this lecture, you should be able to:

- Explain why the DTFT summation is intractable to fully compute, i.e. in computer code/memory, in practice and why this motivates using the DFT.
- Explain the relationship between the DTFT and the DFT, and describe how the two transforms are different.



How can we compute $X_d(\omega)$ in practice?

For a discrete-time signal $x[n]$, consider computing the DTFT $X_d(\omega)$.

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

More specifically, imagine writing a function in your favorite programming language that returns the DTFT of a given input signal.

First issue: We have an infinite-length summation.



How can we compute $X_d(\omega)$ in practice?

In general, we cannot calculate an infinite-length summation in code!

→ Let's approximate $x[n]$ with its first N samples.

→ $\hat{X}_d(\omega) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n} \neq X_d(\omega)$

$= \sum_{n=-\infty}^{\infty} x[n] w[n] e^{-j\omega n}$

$v[n]$

$w[n] = \begin{cases} 1, & 0 \le n \le N-1 \\ 0, & \text{else} \end{cases}$

$= \sum_{n=-\infty}^{\infty} v[n] e^{-j\omega n}$

$= V_d(\omega)$

Recall:

$x[n] w[n] \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} X_d(\omega) * W_d(\omega)$

Thus,

$V_d(\omega) = \hat{X}_d(\omega) = \frac{1}{2\pi} X_d(\omega) * W_d(\omega)$

Second issue:

→ W is continuous

→ $\hat{X}_d(\omega)$ is continuous

How can we compute $X_d(\omega)$ in practice?

Even if $x[n]$ is finite-length, we cannot evaluate infinitely-many frequencies between $[-\pi, \pi]$!

Instead, let's discretize ω:

Let $\omega_k = \frac{2\pi k}{N}$, $0 \le k \le N-1$

$$0 \le \omega_k \le 2\pi \cdot \frac{N-1}{N} < 2\pi$$

$$\hat{X}_k = \sum_{n=0}^{N-1} x[n] e^{-j\omega_k n}, \quad 0 \le k \le N-1$$

$$= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N} n}, \quad 0 \le k \le N-1$$

Discrete Fourier Transform (DFT)



Discrete Fourier transform (DFT)

The discrete Fourier transform of a length–N discrete-time signal is denoted by $X[k]$.

(DFT) $$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n},\ 0 \le k \le N-1$$

(Inverse DFT) $$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n},\ 0 \le n \le N-1$$

- Note that $X[k]$ is defined for $0 \le k \le N-1$!



Relationship between DTFT and DFT

Let $x[n]$ be a length–N discrete-time signal.

$$X_d(\omega) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N} n}$$

Let $\omega_k = \frac{2\pi k}{N} \rightarrow X_d(\omega_k) = \sum_{n=0}^{N-1} x[n] e^{-j\omega_k n} = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N} n}$

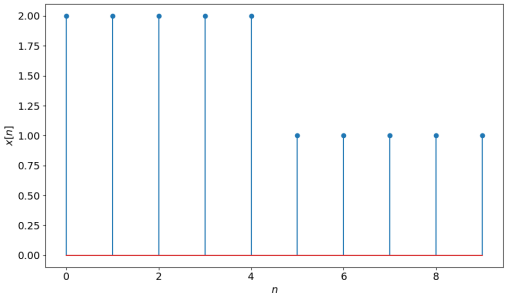
$X[k] = X_d(\omega_k) = X_d\left(\frac{2\pi k}{N}\right) \rightarrow$ The DFT evaluates the DTFT of a length-N signal at N evenly-spaced frequencies for $0 \le \omega_k < 2\pi$.



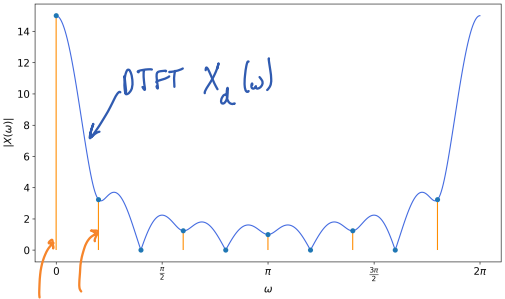
Relationship between DTFT and DFT

$N = 10$

$\omega_k = \frac{2\pi k}{10} = \frac{\pi}{5} k$



(a) Finite-length $x[n]$



(b) $X_d(\omega)$ and values of $X[k]$

N-periodicity of the DFT

For a length–N $x[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]$:

$$X[k] = X[k + mN]$$



N-periodicity of the DFT

For a length–N $x[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]$:

$$X[k] = X[k + mN]$$

Proof:

$$\begin{aligned} X[k+mN] &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi(k+mN)}{N}n} \\ &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n} e^{-j2\pi mn} \\ &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n} \\ &= X[k] \end{aligned}$$



N-periodicity of the DFT

For a length–N $x[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]$:

$$x[n] = x[\mathfrak{n} + mN]$$



N-periodicity of the DFT

For a length–N $x[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]$:

$$x[n] = x[k + mN]$$

Proof:

$$\begin{aligned} x[n + mN] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}(n+mN)} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n} e^{j2\pi km} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n} \\ &= x[n] \end{aligned}$$



How can we get more points from $X_d(\omega)$ in $X[k]$?

What if we want more than N evenly-spaced points from $X_d(\omega)$ evaluated in $X[k]$?



How can we get more points from $X_d(\omega)$ in $X[k]$?

What if we want more than N evenly-spaced points from $X_d(\omega)$ evaluated in $X[k]$?
Let $x[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]$ be a length–N signal. Now, we add L zeros to $x[n]$ to obtain $y[n]$.

$$y[n] = \begin{cases} x[n], & 0 \le n \le N-1 \\ 0, & N \le n \le N+L-1 \end{cases}$$

→ Zero-padding

What is $Y_d(\omega)$ and $Y[k]$?

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=0}^{N-1} x[n] e^{-j\omega n}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N} n}, \quad 0 \le k \le N-1$$

$$X[k] = X_d\left(\frac{2\pi k}{N}\right)$$

$$Y_d(\omega) = \sum_{n=-\infty}^{\infty} y[n] e^{j\omega n}$$
$$= \sum_{n=0}^{N+L-1} y[n] e^{j\omega n}$$
$$= \sum_{n=0}^{N-1} x[n] e^{j\omega n}$$
$$= X_d(\omega)$$

$$Y[k] = \sum_{n=0}^{N+L-1} y[n] e^{-j\frac{2\pi k}{N+L} n}, \quad 0 \le k \le N+L-1$$
$$= \sum_{n=0}^{N-1} y[n] e^{-j\frac{2\pi k}{N+L} n}$$
$$= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N+L} n}$$

$$\rightarrow Y[k] = Y_d\left(\frac{2\pi k}{N+L}\right) = X_d\left(\frac{2\pi k}{N+L}\right)$$

Zero-padding Example 1

Let $x[n]$ be a length–16 signal with DFT $X[k]$. We zero-pad $x[n]$ with 32 zeros to obtain $y[n]$ with DFT $Y[k]$. Which values of $Y[k]$ will be equal to:

(a) $X[0]$
(b) $X[1]$
(c) $X[4]$

$$\omega_k^x = \frac{2\pi k}{N} = \frac{2\pi}{16}k$$

$$\omega_k^y = \frac{2\pi k}{N+L} = \frac{2\pi}{48}k$$

(a) $\omega_0^x = 0$

$\omega_0^y = 0$

$\rightarrow k = 0$

$X[0] = Y[0]$

(b) $\omega_1^x = \frac{2\pi \cdot 1}{16} = \frac{\pi}{8}$

$\omega_k^y = \frac{2\pi k}{48} = \frac{\pi}{24}k$

$\rightarrow \frac{\pi}{24}k = \frac{\pi}{8} \rightarrow k = 3$

$Y[3] = X[1]$

(c) $\omega_4^x = \frac{2\pi \cdot 4}{16} = \frac{\pi}{2}$

$\omega_k^y = \frac{2\pi \cdot k}{48} = \frac{\pi}{24}k$

$\rightarrow \frac{\pi}{24}k = \frac{\pi}{2} \rightarrow k = 12$

$\rightarrow Y[12] = X[4]$

Zero-padding Example 2

Suppose we have a length–N discrete-time signal $x[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)$ and we would like the DFT $X[k]$ to contain $X_d\left(\frac{\pi}{5}\right)$ and $X_d\left(\frac{\pi}{3}\right)$. What are the two smallest possible values of N?

$$\omega_k = \frac{2\pi k}{N}$$

Need N such that: $\omega_{k_1} = \frac{2\pi k_1}{N} = \frac{\pi}{5}$ for some k_1 and k_2.

and

$$\omega_{k_2} = \frac{2\pi k_2}{N} = \frac{\pi}{3}$$

$\rightarrow \frac{2\pi k_1}{N} = \frac{\pi}{5}$ $\rightarrow \frac{2\pi k_2}{N} = \frac{\pi}{3}$ $\Longrightarrow$ N is multiple of 6 and 10

$N = 10k_1$ $N = 6k_2$

$N = 30, 60, 90, \ldots$

Two smallest (Least Common Multiple)